
JVM (Java Virtual Machine)

1 What is JVM?

JVM executes Java bytecode and provides platform independence.

2 What is JRE?

Java Runtime Environment contains libraries + JVM needed to run Java applications.

3 What is JDK?

Java Development Kit includes JRE, compiler, and tools for development.

4 What is ClassLoader?

A component of JVM that loads classes into memory.

5 What is Garbage Collection?

Automatic memory cleanup that removes unused objects.

6 What is JIT Compiler?

Just-In-Time compiler that improves performance by converting bytecode to native code at runtime.

7 JVM Memory Areas?

Heap, Stack, Method Area, PC Registers, Native Method Stack.

8 What is Heap Memory?

Stores objects and shared data used by all threads.

9 What is Stack Memory?

Stores method calls and local variables for individual threads.

10 What are GC Algorithms?

Mark-Sweep, Copy, G1—used to manage and optimize memory usage.

OOPs (Object-Oriented Programming)

1 What is OOPs?

OOPs is a programming paradigm based on real-world entities that organizes code using classes and objects. It increases reusability, flexibility, and security.

2 What is a Class?

A class is a blueprint or template from which objects are created. It defines properties and behaviors.

3 What is an Object?

An object is a real-time entity that represents an instance of a class with its own state and behavior.

4 What is Encapsulation?

Encapsulation means wrapping data and methods into a single unit and restricting direct access using private fields.

5 What is Inheritance?

Inheritance allows one class to acquire properties of another class, enabling code reuse and method overriding.

6 What is Polymorphism?

Polymorphism allows the same method name to behave differently—compile-time (overloading) and runtime (overriding).

7 What is Abstraction?

Abstraction hides implementation details and shows only essential features using abstract classes and interfaces.

8 What is Method Overloading?

Overloading means multiple methods with the same name but different parameters. It improves readability and flexibility.

9 What is Method Overriding?

Overriding means redefining a parent class method in a child class to give a specific implementation.

10 What is the difference between Interface and Abstract Class?

Interfaces provide 100% abstraction by default, while abstract classes can have both abstract and concrete methods.

1 What is Exception Handling?

Exception handling is a mechanism to handle runtime errors so the normal flow of the program continues.

2 What is an Exception?

An exception is an event that interrupts normal program execution at runtime.

3 Difference between Checked and Unchecked Exceptions?

Checked exceptions are checked at compile-time; unchecked exceptions occur at runtime and are not checked.

4 What is the purpose of try-catch?

try contains risky code; catch handles the exception so the program doesn't crash.

5 What is the finally block?

finally executes whether an exception occurs or not—used for resource cleanup.

6 What is throw?

throw is used to explicitly throw an exception inside a method.

7 What is throws?

throws declares that a method may throw certain exceptions and hands responsibility to the caller.

8 Can we use multiple catch blocks?

Yes, multiple catch blocks allow handling of different exception types individually.

9 What is try-with-resources?

A feature that automatically closes resources like streams, avoiding memory leaks.

10 Can we catch multiple exceptions in one catch?

Yes, using `catch(Exception1 | Exception2 e)` introduced in Java 7.

1 What are the main Java 8 features?

Lambda expressions, Stream API, Functional Interfaces, Optional, Default Methods.

2 What is a Lambda Expression?

A short way to write inline code blocks for functional interfaces using `->` syntax.

3 What is Stream API?

A powerful API for processing collections using functional operations like `map`, `filter`, `reduce`.

4 What is Optional?

A wrapper to avoid `NullPointerException` by checking value presence.

5 What is a Functional Interface?

An interface with a single abstract method; e.g., `Runnable`, `Callable`, `Supplier`.

6 What are Default Methods?

Methods in interfaces with a default implementation to avoid breaking old code.

7 What is Method Reference?

A shortcut to refer to existing methods using `::`.

8 What is Collectors?

A utility class used with Stream API to collect results into lists, sets, maps.

9 What is Predicate?

A functional interface that represents a boolean condition used for filtering.

10 Difference between Stream and Collection?

Collections store data; Streams process data in a pipeline without modifying the source.

1 What is the Java Collection Framework?

A set of ready-made data structures and algorithms that simplify data storage, searching, sorting, and manipulation.

2 What is the root interface of Collection Framework?

Collection is the root interface for List, Set, Queue; Map is a separate hierarchy.

3 Difference between List and Set?

List allows duplicates and maintains insertion order; Set does not allow duplicates and may not maintain order.

4 What is ArrayList?

A resizable dynamic array best for fast read operations. It is not synchronized by default.

5 What is LinkedList?

A doubly-linked list good for frequent insert/delete operations but slower for random access.

6 What is HashSet?

A Set implementation backed by a hash table that stores unique elements without guaranteed order.

7 What is LinkedHashSet?

A HashSet with insertion order maintained using a linked list.

8 What is TreeSet?

A sorted set that stores elements in ascending order using a Red-Black Tree.

9 What is HashMap?

A key-value pair collection that allows one null key and multiple null values; uses hashing for fast access.

10 Difference between HashMap and Hashtable?

HashMap is non-synchronized and fast; Hashtable is synchronized and legacy.

1 1 What is ConcurrentHashMap?

A thread-safe map that uses segmented locking for high performance in multithreading.

1 2 Why are HashSet elements unique?

Because it uses hashCode() and equals() to check duplicates.

1 What is a Stream?

A pipeline of data processing operations that supports functional programming.

2 What is Intermediate Operation?

Operations like `map()`, `filter()` that return a new stream without executing immediately.

3 What is Terminal Operation?

Operations like `collect()`, `forEach()`, `reduce()` that trigger actual stream processing.

4 Difference between map and flatMap?

`map` transforms each element; `flatMap` flattens nested structures before processing.

5 What is reduce()?

A terminal operation used to combine elements into a single result (sum, max, etc.).

6 What is parallelStream()?

A multithreaded stream for faster processing on multi-core machines.

7 What is sorted()?

An intermediate operation that sorts elements using natural or custom comparator.

8 What is distinct()?

Removes duplicate elements from a stream based on `equals()`.

9 What is limit() and skip()?

`limit()` reduces stream size; `skip()` ignores initial elements.

10 Can Stream be reused?

No, a Stream can be used only once; reusing causes `IllegalStateException`.

1 What is a Thread?

A thread is a lightweight subprocess that enables parallel execution.

2 How do you create a thread in Java?

By extending Thread or implementing Runnable; both provide the run() method.

3 Difference between start() and run()?

start() creates a new thread, while run() just executes like a normal method.

4 What is Synchronization?

A mechanism to control multiple threads accessing shared resources, preventing race conditions.

5 What is a Race Condition?

A problem where multiple threads access shared data at the same time, causing inconsistent results.

6 What is Deadlock?

A state where two threads wait forever for each other's locks, blocking execution.

7 What is ReentrantLock?

An advanced lock with extra features like fairness, tryLock(), and interruptible wait.

8 What is Thread.sleep()?

It pauses the thread temporarily to free CPU for other tasks.

9 What is wait() & notify()?

Used for inter-thread communication; wait() releases lock, notify() wakes up waiting threads.

10 What is Executor Framework?

A framework that handles thread creation using thread pools, improving performance.